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Understanding urban water performance at the city-region scale using an urban water metabolism evaluation framework

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ABSTRACT

Water sensitive interventions are being promoted to reduce the adverse impacts of urban development on natural water cycles. However it is currently difficult to know the best strategy for their implementation because current and desired urban water performance is not well quantified. This is particularly at the city-region scale, which is important for strategic urban planning. This work aimed to fill this gap by quantifying the water performance of urban systems within city-regions using 'urban water metabolism' evaluation, to inform decisions about water sensitive interventions. To do this we adapted an existing evaluation framework with new methods. In particular, we used land use data for defining system boundaries, and for estimating natural hydrological flows. The criteria for gauging the water performance were water efficiency (in terms of water extracted externally) and hydrological performance (how much natural hydrological flows have changed relative to a nominated pre-urbanised state). We compared these performance criteria for urban systems within three Australian city-regions (South East Queensland, Melbourne and Perth metropolitan areas), under current conditions, and after implementation of example water sensitive interventions (demand management, rainwater/stormwater harvesting, wastewater recycling and increasing perviousness). The respective water efficiencies were found to be 79, 90 and 133 kL/capita/yr. In relation to hydrological performance, stormwater runoff relative to pre-urbanised flows was of most note, estimated to be 2-, 6- and 3- fold, respectively. The estimated performance benefits from water sensitive interventions suggested different priorities for each region, and that combined implementation of a range of interventions may be necessary to make substantive gains in performance. We concluded that the framework is suited to initial screening of the type and scale of water sensitive interventions needed to achieve desired water performance objectives.

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1. Introduction

Innovations in total water cycle management are being promoted to secure water supplies and reduce the impacts of urban development on natural water cycles (ADB, 2016, OECD, 2015, UK Water Partnership, 2015), and articulated in concepts such as 'water-sensitive cities' (Wong and Brown, 2009) and 'water wise cities' (IWA, 2016). These include water resource management interventions such as improved water use efficiency and diversification of supplies (harvesting of rainwater and stormwater runoff, wastewater recycling) (Campisano et al., 2017; Khastagir and Jayasuriya, 2010; Sharma et al., 2013), and planning interventions such as water sensitive urban design (WSUD) and greater green space (Coutts et al., 2013; Feng et al., 2016; Sharma et al., 2016). We collectively refer to these as water-sensitive interventions.

To understand how best to employ these interventions, water managers and urban planners need quantification of urban water performance (i.e., how efficiency water is being utilised, how the natural water cycle is altered) (Chesterfield et al., 2016). They need

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to know what the current urban water performance is, what the desired water performance should be, and the extent to which various interventions can influence performance. Quantification of urban water performance is an evolving science with a range of approaches described (Behzadian and Kapelan, 2015a, b, Makropoulos et al., 2008; Renouf and Kenway, 2016; Rozos and Makropoulos, 2013; Urich et al., 2013; van Leeuwen et al., 2012; Venkatesh et al., 2014).

One approach is 'urban water metabolism' evaluation, which provides a big-picture perspective of urban water performance. It is based on the concept of urban metabolism (Wolman, 1965), which has been operationalised to evaluate material and energy flows through urban areas (examples are Kennedy et al. (2007) and Newman (1999)), and specifically adapted for evaluating urban water performance (Kenway et al., 2011; Renouf and Kenway, 2016). It generates a comprehensive account of all flows of water (natural and anthropogenic) between an urban area and the supporting environment to produce an urban water mass balance, from which indicators of water metabolism performance are derived. These indicators relate to metabolic aspects such as water efficiency, degree of supply internalisation, and the extent to which natural hydrological flow are altered (Renouf et al., 2017).

The approach has been used to characterise the water balances and water performances of cities and metropolitan areas (Bhaskar and Welty, 2012; Kenway et al., 2011; Marteleira et al., 2014; Thériault and Laroche, 2009) and to compare alternative urban water servicing options (Farooqui et al., 2016). The spatial scales of past work has been individual cities or urban areas, with system boundaries dictated by the data available for estimating water flows (Kennedy et al., 2015). Urban system boundaries have previously not been systematically defined, nor specifically aligned to the urban planning context. We propose that a closer alignment with urban planning and water management processes will allow the knowledge generated to better inform decision making (Serrao-Neumann et al., 2017).

In the context of Australia, the 'city-region' is the spatial scale at which strategic urban planning often occurs, especially for fast-growing capital city-regions (Low Choy and Minnery, 1994). City-regions are larger in scale than individual cities, often composed of multiple metropolitan areas, regional centres, and hydrological catchments, and including not only urban areas but also surrounding peri-urban and rural areas. It is proposed that urban water resources should be managed at this scale (Plummer et al., 2011; Vietz et al., 2016).

The wider geographic perspective of the city-region gives water managers and urban planners a broader picture of the implications of water sensitive interventions. For example, a previous study that evaluated alternative water servicing option for a new urban development (Farooqui et al., 2016) observed that the benefits of utilising recycled wastewater outside the urban area (in an adjoining rural area) could not be accounted for because the boundary was constrained to just the urban areas. A broader system boundary that captures the wider influence of water sensitive interventions would help optimise water performance for the city-region overall.

Therefore, the aim of this work was to adapt the urban water metabolism evaluation approach urban systems in the context of the city-region, and to test how it can inform decision about water-sensitive interventions. The specific question asked was: can quantification of urban water metabolism performance in the context of the city-region inform decisions about water sensitive interventions?

The evaluation approach we adapted was based on an urban water mass balance method originally described by Kenway et al. (2011), and further developed into an evaluation framework by Farooqui et al. (2016), using performance indicators proposed by Renouf et al. (2017). Adaption of this approach to the larger urban scale involved defining the urban systems within the city-region, and then estimating the exchanges of water between the urban systems and the supporting city-region. In the absence of an established protocol for defining urban system boundaries (Alberti et al., 2017), it was necessary to develop a new method using land use data. It was also necessary to develop new methods for estimating water flows based on the land use data, so they align to the defined boundary. This meant we fitted the data to the defined boundary, rather than fitting the boundary to the data, which has been the practice in the past.

To answer our research question, the adapted evaluation framework was used to evaluate the water metabolism performance of urban systems within three Australian case study city-regions, to test how this information could inform decisions about urban water management.

The gaps this research fills, and the novel contributions are i) the application of urban water metabolism evaluation at the city-region scale; ii) a systematic method for defining urban system boundaries, and iii) methods for estimating urban water flows to align with a defined urban system boundary.

The work substantially develops urban water metabolism evaluation by providing more rigorous methods to enable consistent comparisons of the water performance of urban areas. It demonstrates how the refined methods can generate water performance information at a scale that aligns with strategic urban and water planning processes. It shows for the first time how this approach can enable a better understanding of the roles of water sensitive interventions in achieving water performance objectives for city-regions, to help manage increasingly water-stressed city-regions.

2. Material and methods

2.1. Method overview

The adapted evaluation framework has four method components: i) definition of the urban system boundary using spatial land use data; ii) estimation of water flows between the urban systems and the city-region; iii) compilation of the water flow data into a water mass balance, and iv) derivation of indicators of urban water metabolism performance from the water mass balance data (Fig. 1). The first and second are new components developed by this research to systematically define the urban system at this large urban scale, and estimate water flows to align with the defined urban system boundary. In the sections that follow, we describe each of these steps.

The evaluation framework was applied to three Australian case study city-regions to characterise and compare their water metabolism performance under current conditions and after implementation of water sensitive interventions, and also under pre-urbanised conditions (as a point of reference). The three case study Australian city-regions were South East Queensland (SEQ), the Melbourne metropolitan region (MEL) and the Perth metropolitan region (PER) (Fig. 2). These are a reasonable representative sample of the different climates, topographies and urban forms present in urban Australia (Table 1). They are examples of increasingly water-stressed city-regions, which face the challenges of planning for future population growth with reduced and more erratic water supply due to climate change (Gooda and Voogt, 2012; Risbey, 2011; Schandl et al., 2008).

2.2. Urban system boundary definition

2.2.1. Spatial boundary of the urban systems within the city-regions

The spatial extents of the case study city-regions were derived

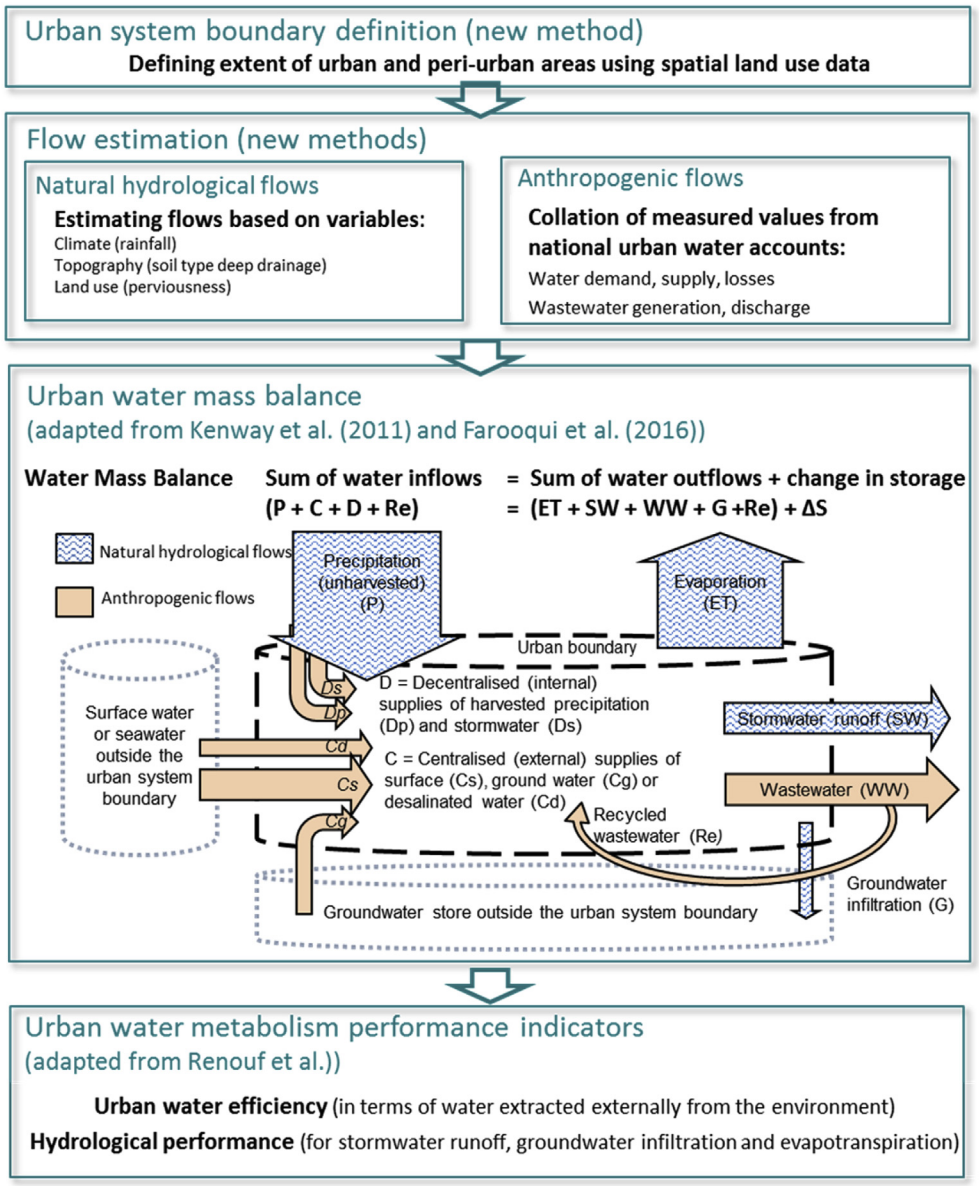


Fig. 1. Urban water metabolism evaluation framework. Based on an urban water mass balance approach originally described by Kenway et al. (2011) and further developed by Farooqui et al. (2016), generating performance indicators proposed by Renouf et al. (2017).

from urban planning documents: the strategic regional plans for growth management (Queensland Government, 2016, Victorian Government, 2014; Western Australian Government, 2015). They are shown in Fig. 2 and in more detail in the Supplementary Material (Figures S1, S2 and S3). Urban water supplies are sourced from the hydrological catchments within the city-regions, so the city-regions also represent the supporting environments from which urban water supplies are drawn.

The 'urban systems' within the city-regions were then defined. We sought to define them in a systematic and repeatable way. Population density was first considered, as used by the OECD (2012) to define urban development. However, because population data can have relatively coarse resolution (being linked to administrative areas), it is difficult to reliably define the spatial extent of urban development (Maheshwari et al., 2016).

The approach we arrived at was to use spatial data to categorise land uses, from which we identified areas that are urban and peri-urban in the nature. Spatial data for land uses are available at very high level of resolution (100 m^2) in geographical information systems (GIS). This allowed detailed spatial statistical analysis using ArcGIS 10.3 software. The GIS land use data for the case study regions for the year 2014 was sourced from the Catchment Scale Land Use of Australia 2015 database (CLUM 2015), produced by the Australian Bureau of Agricultural and Resource Economics (ABARE, 2015), and categorised using the Australian Land Use and Management Classifications (ALUM) system (ABARE, 2013). The land use information can be linked to imperviousness factors for estimating natural hydrological flows in subsequent steps (see Section 2.3.2).

A limitation of using spatial land use data is that the distribution of land uses at the periphery of urban areas is very fragmented, making it difficult to define an actual boundary 'envelope' in a consistent and reproducible way. To address this, 'urban footprints' defined in strategic regional plans were also considered. Urban footprints define the agreed extent of urban development for a strategic planning horizon (around 25 years). It provides an agreed

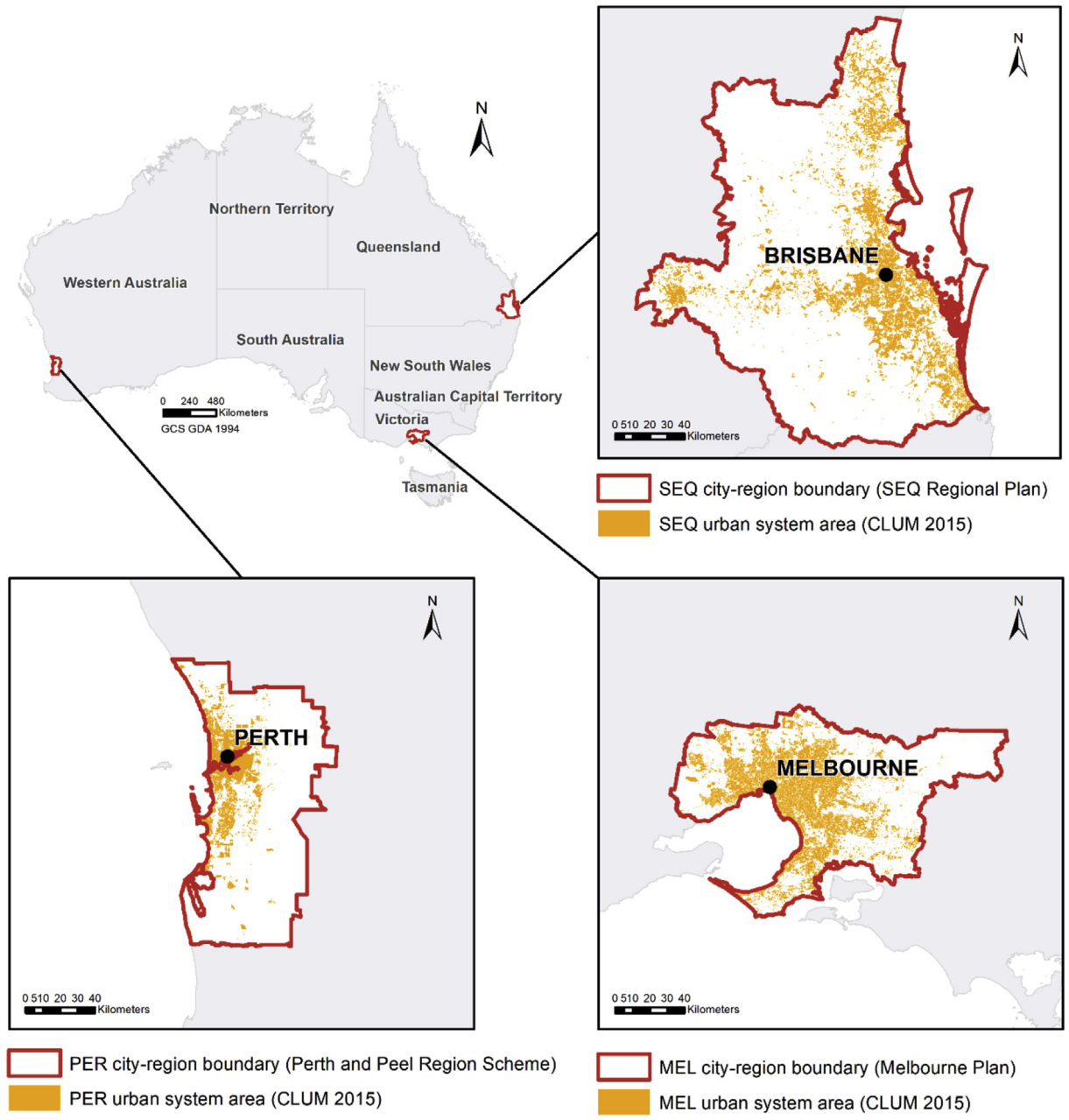


Fig. 2. Location of the Case study Australian city-regions and the defined urban system areas within them. The boundary lines show the extent of the city-regions as described in strategic regional plans. The shading within the boundary line shows the urban system areas, comprising urban and peri-urban land uses (as at 2014). CLUM 2015 refers to the Catchment Scale Landuse of Australia 2015 database.

Table 1
Features of the case study urban systems.

	South East Queensland (SEQ)	Melbourne (MEL)	Perth (PER)
Area of urban systems within the city-region (km^2)	4318	2879	955
Climate	Sub-tropical/wet Heavy summer rain	Temperate/wet Rainfall distributed over the year	Temperate/very dry Very dry summer
Rainfall (2013/14) [1]	681 mm	845 mm	771 mm
Major water sources (2013/14)	Surface water	Surface water	Groundwater Surface water, Seawater desalination

Notes: 1. From Australian Bureau of Meteorology (BOM, 2014).

boundary for what is regarded by the planning sector as urban (and future urban), and a useful envelope for analysing land use with GIS. The urban footprints for the case study city-regions are shown in the Supplementary Material (Figs. S1, S2 and S3).

We identified land uses within the urban footprints that can be considered urban in nature (see the Supplementary Material),

which broadly include residential, commercial, industrial, recreational, cultural and natural areas. Peri-urban areas (those categorised as rural residential) were also included as part of the urban system. They are the lower-density fringe on the periphery of urban areas, and are present both within and outside the urban footprtint perimeter. They were included because they are a significant feature of Australian city-regions, and many water sensitive interventions (such as stormwater harvesting and wastewater recycling) will occur in or influence these areas due to open space availability.

The resulting urban system areas identified within each of the city-regions are shown in Fig. 2, and in more detail in the Supplementary Material (Figs. S1, S2 and S3). They represent the system boundaries for the analyses, and the calculated land areas of the urban systems are shown in Table 1. Vertically, the urban system extends from rooftop and tree tops, to the root zones of trees (assumed to be 1 m below ground) (Kenway et al., 2011).

2.2.2. Technical boundary to distinguish between 'external' and 'internal' water sources

Our interpretation of urban water metabolism suggests a desire to reduce reliance on freshwater extracted externally from the environment, and so the indicator of water efficiency generated by the framework is in terms of water extracted externally (discussed in Section 2.5). Therefore it was necessary to distinguish between water sourced from the environment ('external' water) and water sourced from within the urban system ('internal' water) (Renouf et al., 2017). This technical boundary definition is analogous with technical system boundary definition well-established in environmental life cycle assessment for production systems (Rebitzer et al., 2004).

Centralised water supplies sourced from surface or ground waters are regarded as external water sources, because they originate from environment stocks. Desalinated water was also categorised as external water, because it originates from seawater, even though the freshwater is generated within the urban system. In comparison, decentralised supplies from rainwater, stormwater and recycled wastewater are harvested from within the urban system, and are regarded as 'internal' water sources. The terms 'centralised' and 'decentralised' do not adequately describe this distinction, as they refer to the way water supplies are managed rather than where they come from. However, we have retained the use of these terms to be consistent with industry terminology, and clarify our meaning by calling them centralised (external) water versus decentralised (internal) water.

Groundwater would generally be considered an external water source. In the PER case however, groundwater from the region's very high water tables is an important urban water supply, and infrastructure is being built to use aquifers as water storage reservoirs with managed recharge, including injection of highly-treated wastewater (Water Corporation n.d.). Aquifers that function as such could be regarded as internal water sources. However, we categorised water from Perth's groundwater aquifers as external water for consistency.

2.3. Estimation of water flows

The water flows estimated for the analyses are exchanges of water between the urban system areas and the supporting city-region, as annually averaged flows (GL/yr). The nomenclature to describe individual flows (see Fig. 1) was based on the original evaluation framework of Farooqui et al. (2016), but with the further refinements of Renouf et al. (2017) to differentiate between water sourced internally and externally (see Section 2.2.2). In order for the water flow data to align with the defined urban system boundaries, new estimation methods and data sources were used.

2.3.1. Estimation of anthropogenic flows

Anthropogenic flows are those managed by urban water infrastructure (orange arrows in Fig. 1), i.e., centralised (C) and decentralised (D) water supplies, and wastewater discharged (WW) or recycled (Re). Data for the centralised anthropogenic flows were mostly sourced from the Australian Bureau of Meteorology's Urban Water Accounts (BOM UWA) database (BOM, 2016), which generates Australia's National Water Accounts (BOM, 2015). It compiles annual volumes of water supplied by bulk water providers and retailers (potable and non-potable to residential, commercial and industrial), and wastewater generated, treated and recycled by wastewater treatment utilities. The data derived from the database for the year of the study (2013/2014) was aligned to the defined urban system boundaries by only including data for the water utilities that service these areas, which are described in the following sections.

The BOM UWA database does not report decentralised supplies from harvested rainwater, stormwater or bore water. These were instead derived from published reports and surveys (Table 2). Harvested rainwater was estimated from the amount of precipitation, average household roof area, number of households, tank volumetric reliability (Umapathi et al., 2012), and tank penetration (ABS, 2013). Data was not available for dam and bore water collected or extracted in peri-urban areas not serviced by the potable urban water supply.

2.3.2. Estimation of natural hydrological flows

Natural hydrological flows are those in the natural water cycle (blue arrows in Fig. 1), i.e., precipitation (P), stormwater runoff (SW), infiltration to groundwater (G), and evapotranspiration (ET). Water that flows through the urban system in natural watercourses (creeks, rivers, aquifers etc.) is not accounted for as it is considered part of the environment.

To estimate the size of SW, G and ET we first made assumptions about the fraction of P that become SW, G, and ET, which we termed hydrological flow partitioning (HFP) factors. These were derived from runoff and infiltration coefficients using the U.S. Soil Conservation Service runoff curve number method (USDA Soil Conservation Service, 1986), and the Australian Bureau of Meteorology's (BOM) Australian Landscape Water Balance database (BOM, 2017), respectively. Runoff coefficients refer to the fraction of P that becomes SW. The runoff curve number method used to derive them is based on soil type, land use type (imperviousness) and daily precipitation. After deducting the estimated SW fraction from P, the remainder was assumed to either evapotranspire (ET) or infiltrate (G). The amount that infiltrates (G) was estimated using the average infiltration coefficient for each city region, derived from the amount of deep drainage reported in the BOM landscape water balance. The remainder was then assumed to evapotranspire (ET). The resulting HFP factors for each land use type in each city-region are reported in Table 3, and further details of their derivation are included in the Supplementary Material.

To estimate the total annual volumes of SW, G and ET (GL/yr), the HFP factors for each land use type (Table 3) were multiplied by the areas of the respective land use types within the urban system.

2.4. Urban water mass balance

The estimates of anthropogenic and natural flows (GL/yr) were brought together into the urban water mass balance (equations (1) and (2) below), using the method described in the original framework of Farooqui et al. (2016). The aim was to achieve a mass balance, such that total inflows equal total outflows, plus any changes

Table 2
Summary of data sources.

Data	Source	Year of data
Land use data for defining urban systems	Catchment Scale Land Use of Australia 2015 database (CLUM 2015) Version 7 (ABARE, 2015), using the Australian Land Use and Management Classifications (ALUM) system (ABARE, 2013)	2014
Anthropogenic water flows (centralised)		
- Cs, Cg, Cd - WW - Re	Australian Bureau of Meteorology's (BOM) Urban Water Accounts database (BOM, 2016) - data for the water utility jurisdictions within the defined urban system area were extracted from the database	2013–2014
Anthropogenic water flows (decentralised)		
- Dp	Published survey of the extent of rainwater harvesting in the city-regions (ABS, 2013), from which average rates of harvesting were derived.	2013
- Ds	Published estimates (Whiteoak et al., 2012)	2012
- Dg	Published estimates (Government of Western Australia n.d.)	
Natural hydrological flows		
- P	Total area-averaged annual rainfall database (BOM, 2014) for the defined urban system areas. It is compiled by BOM from weather stations within the respective regions.	2013–2014
- SW	Coefficients derived from the U.S. Soil Conservation Service runoff curve number method (USDA Soil Conservation Service, 1986)	2013–2014
- G	Coefficients derived from the annual deep drainage reported in the BOM Australian Landscape Water Balance (BOM, 2017)	2013–2014
- ET	Remainder after deducting runoff and infiltration	2013–2014
Urban population	Australian Bureau of Statistics (ABS, 2015) – populations for the statistical local areas within the urban system area.	2013–2014

Note: Water flow acronyms are described in Section 2.4.

Table 3
Hydrological flow partitioning (HFP) factors.

Land use category	SEQ			MEL			PER		
	ET	G	SW	ET	G	SW	ET	G	SW
Urban residential	0.66	0.03	0.31	0.86	0.05	0.09	0.78	0.17	0.05
Manufacturing	0.64	0.03	0.33	0.84	0.05	0.11	0.76	0.16	0.08
Commercial and public services	0.55	0.02	0.42	0.76	0.04	0.19	0.66	0.14	0.19
Roads, airports, transport and communications	0.34	0.01	0.65	0.50	0.03	0.47	0.30	0.07	0.63
Utilities and municipal services	0.73	0.03	0.24	0.90	0.05	0.04	0.81	0.18	0.02
Mining, extractive industries	0.73	0.03	0.24	0.90	0.05	0.04	0.81	0.18	0.02
Recreation and cultural	0.85	0.04	0.12	0.93	0.05	0.02	0.81	0.18	0.01
Natural environment	0.87	0.04	0.09	0.93	0.05	0.02	0.81	0.18	0.01
Rural residential	0.83	0.04	0.14	0.93	0.05	0.02	0.81	0.18	0.01

Note: Totals may not add up to one due to rounding. Details of estimating these HFP factors can be found in the Supplementary Material.

in storage, thereby ensuring a comprehensive and accurate account (equations (1) and (2)). Changes in storage were assumed to be zero. In this context storage refers to soil moisture and water stored in reservoirs within the urban system. So no changes in storage assumes stable climatic conditions and no influence on this stored water.

$$\text{Input (Qi)} = \text{Output (Qo)} + \Delta S \quad (1)$$

$$(P + C + D + Re) = (ET + SW + WW + G + Re) + \Delta S \quad (2)$$

Where:

P = unharvested precipitation falling in the urban boundary, i.e. total precipitation less any rainwater or stormwater harvested within the urban system area
C = total centralised (external) water supplies, which includes surface waters (Cs), groundwater (Cg) and desalinated water (Cd)
D = total decentralised (internal) water supplies harvested from within the urban system area, which includes harvested precipitation (rainwater) (Dp) and harvested surface water runoff (Ds), and bore water (Dg).
ET = evapotranspiration from the urban system areas
SW = runoff of surface water/stormwater discharged from the urban system areas (not including that which is harvested)
WW = wastewater discharged from the urban system areas (total wastewater generated less that which is recycled)
G = infiltration to groundwater
Re = reuse/recycling of wastewater
ΔS = change in the stored water within the defined urban system

2.5. Urban water metabolism performance indicators

The following urban water metabolism performance indicators were derived from the water mass balance data using the equations in Table 4. They were based on a set of indicators proposed by Renouf et al. (2017) and previously demonstrated by Farooqui et al. (2016).

Urban water efficiency is an indicator of the overall efficiency of water use for the defined urban system, in terms of water sourced externally from outside the urban system. It is expressed as a rate of external water withdrawal per inhabitant per year (kL/capita/yr). It is an outcome of both the urban water demand, and the degree of supply internalisation (rainwater and stormwater collection, or wastewater recycling).

Hydrological performance is an indicator of the degree to

Table 4
Urban water metabolism indicators.

Indicator	Description	Equation
Urban water efficiency	Total external water use per capita per year (kL/capita/yr)	$\frac{C}{Population}$
Water supply internalisation	Proportion of total urban water demand met by internally harvested/recycled water	$\frac{D+Re}{D+Re+C}$
Hydrological performance	Ratio of post- (i) to pre-urbanised (o) annual flows of stormwater runoff (SW), evapotranspiration (ET, and groundwater infiltration (G)	$\frac{SWi}{SWo}$, $\frac{Gi}{Go}$, $\frac{ETi}{ETo}$

Note: Water flow acronyms are described in Section 2.4.

which natural hydrological flows have increased or decreased relative to a nominated pre-urbanised reference state. It is the ratio of post-to pre-urbanised annual flows for stormwater runoff, evapotranspiration, and infiltration to groundwater. A ratio of greater than one means that the magnitude of the annual flow is larger than the pre-developed landscape, and a ratio of less than one means it is smaller. The pre-urbanised state provides a useful point of reference for gauging performance and tracking changes. Hydrological flows for the pre-urbanised reference were approximated by using the HFP factors for the conservation and natural environment land use category.

2.6. Case studies of water sensitive interventions

The current case, against which water sensitive interventions were compared, represents current conditions using data for 2013/14. The water sensitive (WS) interventions represent a selection of initiatives being promoted in Australia, i.e., reducing water demand, internal harvesting of water (rainwater, stormwater), recycling wastewater, and increasing perviousness to reduce runoff and increase groundwater infiltration (Brown and Farrelly, 2009). The scale of the interventions were based on what might be reasonably achievable in the context of the city-region. For example, in SEQ and MEL there is capacity for harvesting rainwater/stormwater and recycling wastewater. However, for PER harvesting of rainwater and stormwater is less feasible due to the limited rainfall and very high perviousness due to sandy soils. In addition, PER has a high proportion of decentralised use of groundwater in local bores, which currently functions in a similar way to rainwater tanks for outdoor water use. Therefore there is no internal harvesting scenario for PER. The mass balances and water metabolism indicators for these interventions were generated by changing the relevant variables (Table 5).

- **WS1: Reduced demand scenarios** were based on reducing demand for centralised water supply in SEQ, MEL and PER by 8%, 18% and 15%. The demand reductions that had previously occurred in SEQ were used as a benchmark for demand reductions in the other city-regions.
- **WS2: Internal harvesting scenarios** were based on a combination of rainwater harvesting and stormwater harvesting in the SEQ and MEL cases only. Rainwater harvesting is based on an increased percentage of houses with a rainwater tank installed, from 47% to 60% in SEQ and from 31% to 40% in MEL. Stormwater harvesting is based on an increased percentage of harvestable stormwater that is harvested from 0.04% to 5% in SEQ and from 0.3% to 5% in MEL. Rainwater and stormwater harvesting combined results in a reduction of centralised water supply by 12%, and 9%, respectively.
- **WS3: Wastewater recycling scenarios** were based on increasing the percentage of wastewater that is recycled from 7% to 20% in SEQ, from 10% to 20% in MEL, and from 6% to 40% in PER. This results in a reduction of centralised water supply by 11%, 9% and 10%, respectively.
- **WS4: Increasing perviousness scenarios** were based on increasing the perviousness in the urban residential areas to be similar to that of rural residential areas.

Pre-urbanised conditions were also used as the point of reference for generating the indicators of hydrological performance. and were assumed to be the natural landscape with hydrological flows based on the HFP factors for the natural environment land use category.

3. Results

The water mass balance results (Table 6) summarise our estimates of all water inflows and outflows for the water sensitive (WS) interventions, relative to the current (2013/14) and pre-urbanised cases.

For SEQ and MEL, the majority of anthropogenic water inflow is potable water sourced from surface waters, and supplied via the centralised distribution systems. They also have small amounts of decentralised water (from rainwater and stormwater, or from wastewater recycling). For PER, the major anthropogenic inflow is centralised potable water from groundwater and seawater desalination, but also from domestic bores. Compared to the pre-urbanised states, there has been an increase in the total flow of water (anthropogenic and natural) passing through the urban systems (Qi and Qo) by 10% for SEQ, 20% for MEL and 49% for PER (Table 6). The noticeably higher change for PER can be attributed to PER having a relatively smaller area compared to SEQ and MEL, which distorts the ratio between rainfall and anthropogenic inflows.

3.1. Water metabolism characteristics

The results for the water metabolism indicators (Fig. 3) provide a more meaningful interpretation of the water mass balance results. The urban water efficiency results (Fig. 3a–c) combine the total urban water demand (the height of the bars), the amount of internal water supply (the top portion), and the net use of external water (the bottom portion). The hydrological performance indicators (Fig. 4d–f) show how much the current annual volumes of stormwater runoff, evapotranspiration and infiltration have changed compared to the estimated pre-urbanised hydrology.

3.1.1. South East Queensland (SEQ)

SEQ currently has a relatively low total water demand (92 kL/capita/yr) compared to the other city-regions (Fig. 3a). This is in part due to behaviour change embedded during a preceding drought when severe water restrictions and successful water conservation campaigns were imposed (Walton and Hume, 2011). 14% of supply is currently sourced from internal supplies (from wastewater recycling, and relatively small amounts of rainwater and stormwater harvesting), making the amount sourced externally 79 kL/capita/yr. Relative to pre-urbanised hydrological flows, surface water runoff was estimated to be 201%, and groundwater

Table 5
Water sensitive interventions.

Scenarios		Variables				
Region	Water sensitive interventions	Urban water demand (kL/p/yr)	Percentage of households with a rainwater tank installed	Percentage of harvestable stormwater[a] that is harvested	Percentage of wastewater recycled	Hydrological flow partitioning factors of the urban residential areas (ET:G:SW)
SEQ	Pre-urbanised	NA	NA	NA	NA	0.87: 0.04: 0.09[b]
	Current (2013/14)	92	47%	0.04%	7%	0.66: 0.03: 0.31
	WS1 Reduced demand	85	–	–	–	–
	WS2 Internal harvesting	–	60%	5%	–	–
	WS3 Wastewater recycling	–	–	–	20%	–
	WS4 Increase perviousness	–	–	–	–	0.77: 0.03: 0.20
MEL	Pre-urbanised	NA	NA	NA	NA	0.93: 0.05: 0.02[b]
	Current (2013/14)	107	31%	0.30%	10%	0.86: 0.05: 0.09
	WS1 Reduced demand	91	–	–	–	–
	WS2 Internal harvesting	–	40%	5%	–	–
	WS3 Wastewater recycling	–	–	–	20%	–
	WS4 Increase perviousness	–	–	–	–	0.92: 0.05: 0.02
PER	Pre-urbanised	NA	NA	NA	NA	0.81: 0.18: 0.01[b]
	Current (2013/14)	182	9%	0.00%	6%	0.78: 0.17: 0.05
	WS1 Reduced demand	163	–	–	–	–
	WS2 Internal harvesting	–	–	–	–	–
	WS3 Wastewater recycling	–	–	–	40%	–
	WS4 Increase perviousness	–	–	–	–	0.92: 0.05: 0.02

[a] Harvestable stormwater potential is the amount of stormwater runoff that could be harvested without compromising natural hydrological flows, i.e., total estimated runoff (from the urbanised state) minus estimated runoff from the pre-urbanised state.
[b] The hydrological partitioning factors (HFP) for the pre-urbanised case were assumed to be that of the natural environment.

infiltration was 92% and evapotranspiration was 91% (Fig. 3b).

Of the interventions assessed, internal harvesting of rainwater/stormwater (WS2) and wastewater recycling (WS3) may have the greatest influence on the extraction of external water, with both reducing it from 79 to around 70 kL/capita/yr. In comparison, residential demand reduction, which is quite modest (8%) due to SEQ's earlier demand management successes, reduce external water use to 72 kL/capita/yr. If all of these interventions were combined, it would reduce to 54 kL/capita/yr. Increasing perviousness (WS4) to the degree analysed was predicted to reduce surface water runoff by 12%. The increase in groundwater infiltration and evapotranspiration from increased perviousness would be quite minor, increasing groundwater infiltration from 92% to 95%, and evapotranspiration from 91% to 94%, relative to the nominated pre-urbanised state.

3.1.2. Melbourne (MEL)

Total water demand in MEL is 107 kL/capita/yr, and 16% is supplied is from internal water sources, making the amount of water sourced from the environment 90 kL/capita/yr (Fig. 3c). In relation to hydrological flows, surface water runoff was estimated to be 584% of pre-urbanised flows, groundwater infiltration 95% and evapotranspiration 96% of the pre-urbanised state (Fig. 3d). This high increase in runoff is due to MEL having higher urban density and high levels of imperviousness compared with the other city-regions.

Demand management (WS1) by the extent assessed here (18%) could lead to the greatest influence on external water efficiency, reducing it from 90 to 74 kL/capita/yr. In comparison, greater internalisation of water supply (WS2) and wastewater recycling (WS3) each reduces overall environment water use to 82 kL/capita/yr. If all of these interventions were combined, it would reduce to 58 kL/capita/yr. Increasing perviousness to the degree analysed was predicted to reduce surface water runoff from 584% to 472% of pre-urbanised flows. Relative groundwater infiltration would increase from 95% to 97%, and evapotranspiration from 96% to 98%, relative to the nominated pre-urbanised state.

3.1.3. Perth (PER)

PER has a much higher water demand than the other city-regions at around 182 kL/capita/yr. This has been attributed to access to alternative water sources (desalination and household bores) which has reduced the extent of household water conservation (Lam et al., 2016; Lindsay et al., 2017). 27% is supplied from internal water sources, resulting in an overall rate of externally sourced water use of 133 kL/capita/yr (Fig. 3e). In this analysis, the relatively high level of desalinated water inputs was regarded as an external source.

In relation to hydrological flows, surface water runoff was estimated to be 336% of the pre-urbanised state (Fig. 3f). The hydrological impact of highly permeable soil in this region was discussed by Barron et al. (2013). Their work shows a reduction in evapotranspiration and an increase in runoff after urbanisation. In our work, it is interesting to note that evapotranspiration was predicted to increase relative to the pre-urbanised state, rather than decrease. This difference is mainly because in our water balance, we considered anthropogenic water flows that includes high outdoor water use for irrigation of lawns and gardens (Bhaskar et al., 2016).

The scale of demand management assessed here (15%) (WS1) could reduce extraction of external water, from 133 to 113 kL/capita/yr. Water recycling has a similar effect, reducing it to 119 kL/capita/yr. If both of these interventions were combined, it would

Table 6
Urban water mass balance results (annual flows in GL/yr). Flows that changed relative to the current case are shaded.

	Water demand	Inflows								Outflows				Re	Mass Balance		
	C + D + Re	P	C	(C_s)	(C_g)	(C_d)	D	(D_r)	(D_s)	ET	SW	G	WW		Qi	Qo	ΔS
SEQ urban system																	
Pre-urbanised	–	2804	–	–	–	–	–	–	-	2397	298	109	–	–	2804	2804	0
Current (2013/14)	304	2790	261	(260)	(1)	(0.2)	14	(13.9)	(0.2)	2184	599	100	182	29	3094	3094	0
WS1-Reduced demand	283	2790	240	(239)	(1)	(0.2)	14	(13.9)	(0.2)	2184	599	100	161	29	3073	3073	0
WS2-Internal harvest	304	2758	229	(227)	(1)	(0.2)	46	(17.7)	(28.5)	2181	570	100	182	29	3062	3062	0
WS3-Recycling	304	2790	233	(232)	(1)	(0.2)	14	(13.9)	(0.2)	2184	599	100	154	57	3094	3094	0
WS4-Increased perviousness	304	2790	261	(260)	(1)	(0.2)	14	(13.9)	(0.2)	2255	525	104	182	29	3094	3094	0
MEL urban system																	
Pre-urbanised	–	2433	–	–	–	–	–	–	–	2253	49	130	–	–	2433	2433	0
Current (2013/14)	494	2415	415	(415)	(0)	(0)	18	(16)	(2)	2153	286	124	285	61	2908	2908	0
WS1-Reduced demand	421	2415	342	(342)	(0)	(0)	18	(16)	(2)	2153	286	124	212	61	2836	2836	0
WS2-Internal harvest	494	2377	377	(377)	(0)	(0)	55	(20)	(35)	2149	253	123	285	61	2871	2871	0
WS3-Recycling	494	2415	377	(377)	(0)	(0)	18	(16)	(2)	2153	286	124	247	99	2908	2908	0
WS4-Increased perviousness	494	2415	415	(415)	(0)	(0)	18	(16)	(2)	2205	232	127	285	61	2908	2908	0
PER urban system																	
Pre-urbanised	–	742	–	–	–	–	–	–	–	601	10	131	–	–	742	742	0
Current (2013/14)	369	740	268	(30)	(125)	(113)	90	(90)	(0)	766	35	167	131	10	1109	1109	0
WS1-Reduced demand	329	740	228	(30)	(125)	(73)	90	(90)	(0)	766	35	167	91	10	1069	1069	0
WS2-Internal harvest	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
WS3-Recycling	369	740	240	(30)	(125)	(85)	90	(90)	(0)	766	35	167	103	38	1109	1109	0
WS4-Increased perviousness	369	740	268	(30)	(125)	(113)	90	(90)	(0)	784	13	171	131	10	1109	1109	0

Notes: City-regions: SEQ = South East Queensland; MEL = Melbourne; PER=Perth.
Scenarios: Pre-urbanised = conditions assumed to be in place prior to urbanisation, i.e. a natural landscape; Current = approximation of current conditions based on 2013/14 data; WS = hypothetical water sensitive interventions, (see Table 5).
Acronyms for water flows: P = unharvested precipitation, C = total centralised (external) water supply, Cs = centralised (external) supply from surface waters, Cg = centralised (external) supply from groundwater, Cd = centralised (external) supply of desalinated water, D = total de-centralised (internal) water supply, Dp = de-centralised (internal) supply from harvested precipitation (rainwater), Ds = de-centralised (internal) supply from harvested stormwater, ET = evapotranspiration, SW = stormwater runoff, G = groundwater infiltration, WW = wastewater discharged, Re = recycled wastewater, Qi = total inflows (P+C+D), Qo = total outflows (ET+SW+G+WW), ΔS = change in storage.

reduce to around 100 kL/capita/yr. Increasing perviousness to the degree analysed was predicted to reduce runoff from 336% to 125% of pre-urbanised flows. This is considerably higher than observed for the SEQ and MEL cases because PER's baseline runoff is smaller due to relatively high perviousness and sandy soil type. So the relative improvement is amplified. Groundwater infiltration is expected to increase from 96% to 99%, and evapotranspiration from 116% to 118%, relative to the nominated pre-urbanised state.

4. Discussion

The urban water metabolism evaluation of the three Australian city-regions provided a broad picture of current urban water performance, highlighting differences between the city-regions and suggesting different improvement priorities.

For example, the urban systems within the SEQ city-region have relatively low per capita water demand, but considerable scope for improving the internalisation of water supply to reduce reliance on external water. Total stormwater runoff, relative to the pre-urbanised reference state, also appears to be low for SEQ compared with the other city-regions. What is hidden in the SEQ results is the temporal characteristics of runoff, which due to the sub-tropical rainfall patterns would be concentrated in more intense runoff events (discussed further in the limitation section). For urban systems in the MEL city-region, there is scope for reducing demand (if water demand in SEQ is used as a benchmark) and for improving internalisation of supply. A stand-out observation for MEL is the high stormwater runoff relative to the nominated pre-urbanised state, suggesting that managing the overall volumes of runoff may be a priority for MEL. Urban systems within the PER city-region have a higher internalisation of water supply than the other city-regions, but reducing water demand would appear to be a priority.

These water performance observations are already understood by water managers in these regions. What the evaluation adds is the ability to quantify performance to align with the spatial scale of urban planning and regional water management processes. The quantified water metabolism performance indicators can be used to make consistent comparisons between city-regions for benchmarking, track changes over time, and identify broad opportunities for improvement. Furthermore they can promote dialogue across stakeholder groups about what the desired performance for the future should look like.

The value of presenting the hydrological performance indicators as relative values (normalised against the pre-urbanised reference state) rather than as absolute values warrants discussion. This indicator is a 'distance to target' type of indicator, which judges performance in terms of how far the hydrological flows are from a regionally-relevant reference point rather than the actual scale of the flows. This allows for region-specific reference state to be taken into consideration when judging and comparing performance. While it is recognised that restoring natural hydrological flow to the pre-urbanised state may not be the end objective, it does provide a useful point of reference that takes account of regional characteristics. For comparing the urban water performance of the city-regions and the water sensitive interventions, the relative values enabled a more valid comparison would be possible if the results were presented as absolute flows.

Evaluating water sensitive interventions at this larger urban scale enabled the relative significance of the individual or combined interventions to be understood. For example, each of the water sensitive interventions may be considered potentially significant in their own rights, but when considered in the context of the city-region, the benefits may actually be quite modest. Increasing perviousness to the scale assessed here appears to reduce runoff only marginally at the large scale in SEQ and MEL (Fig. 3b,d). Combined implementation of a number of interventions would be necessary to make substantive gains in the overall water



Fig. 3. Urban water metabolism indicators. (WS 1: reduced demand; WS 2: internal harvesting; WS 3: wastewater recycling; WS 4: increase perviousness).

metabolism performance. Individual initiatives may not have much influence in isolation.

A novel contribution was the development of methods for defining the urban system boundary and for estimating hydrological flows using land use data. This enabled the water flow data to be generated in systematic, consistent and repeatable way, so that valid comparisons can be made between different urban systems. A previous water mass balance study by Kenway et al. (2011) of the same regions in 2004/2005 generated urban water flow estimates that were quite different to those estimated in this study for 2013/14. This can be mostly attributed to different urban system boundaries (areas were 1281 versus 4118 km^2 for SEQ, 1818 versus 2879 km^2 for MEL and 1200 versus 955 km^2 for PER). The system boundaries of the Kenway et al. (2011) study were defined by the available data in each region, and may not have represented comparable systems. The more systematic methods used in this work means that the systems are comparable in terms of the urban entity they represent and the flow data generated.

The water metabolism performance indicators generated could be reported alongside other urban water performance indicators (Behzadian and Kapelan, 2015a, CRC WSC, 2016; van Leeuwen et al., 2012; Venkatesh et al., 2014). In particular, the Water Sensitive Cities (WSC) Index (CRC WSC, 2016) provides a framework for reporting urban water performance across biophysical, social and governance aspects. The water metabolism performance indicators demonstrated here could fill a current gap in the quantification of resource management performance in the WSC Index. Reporting the water metabolism indicators within a broader framework also allows other factors, such as public acceptance (Campisano et al., 2017; West et al., 2016) to be considered when assessing water sensitive interventions.

4.1. Limitations

The estimation of flows is a limitation of the method as it compressed the temporal variations of water flows (eg daily, monthly variation) into a single annual total. This means that the temporal characteristics of runoff is hidden, as noted earlier for the SEQ case. Consequently, the method provides a 'snapshot' for a given year and for the urban system at that point in time. To provide

more detailed evaluation of water sensitive interventions, including consideration of storage of water within cities, drawing on data from multiple years, and/or analysis of a finer time-scale (for eg. monthly) is recommended in order to capture the temporal effects.

Similarly, spatial variability of rainfall was not considered. Average rainfall across the assessed urban systems was assumed, but in reality variations in rainfall across this scale could be substantial (perhaps in the order of 200 mm average rainfall across the city-region). This could be addressed in future iterations by partitioning urban systems into representative zones perhaps based on rainfall districts, local government areas, or catchments.

Another limitation is related to the hydrological flow estimation based on land use types and their respective hydrological flow partitioning (HFP) factors. They were sufficient for testing the functionality of the framework, but whether these coarse averages are accurate enough for the purpose of screening and prioritisation of interventions still needs to be validated. The feasibility and preferred location for promising interventions would require more detailed hydrological modelling techniques. In future work, we expect the need to estimate hydrological flows for sub-regions using hydrological models.

4.2. Future work

In this work, we examined a limited set of variables related to water resource management (water demand, water sources, and perviousness) to prove the concept. Other variables related to planning interventions could also be evaluated using the framework. For instance, estimating the influence of population growth, housing densities, land use distributions, and green spaces would help evaluate planning alternatives in the context of water resources.

Climate change is also an important variable to consider in the future applications of the framework. It can lead to temporal redistribution of precipitation (Franczyk and Chang, 2009), and change in rainfall intensities and runoff peak flow (Zahmatkesh et al., 2015), which influence the benefits of water sensitive interventions. However as noted earlier, temporal and spatial variabilities in water flows are not observable with the framework, and so it may not be the best method for considering climate change influences.

The capacity of the evaluation framework for informing urban water management could be expanded by considering the other water-related resource efficiency aspects of energy and nutrients, and functionality of water in the urban landscape (Renouf et al., 2017). Only efficiency and hydrological performance was considered here as a first step to understanding water metabolism performance at this scale. In the case of PER, water-related energy is an important consideration given the high use of energy-intensive desalinated water (Fig. 3e), and its consideration would amplify the benefits of initiatives that displace desalinated water. For SEQ and MEL, the benefits of internal harvesting of rainwater/stormwater appear similar to those from wastewater recycling (Fig. 3 a,c). However, if the nutrient recovery benefits of wastewater recycling were considered, wastewater may be more beneficial. The mass balance framework provides a good foundation on which to add a water-related energy or water-related nutrient analysis due to its systematic characterisation of all flows.

In summary, research needed to further develop the approach would include i) being able to capture the temporal and spatial variation in water flows, ii) validating the HFP factors used to estimate hydrological flows at larger urban scales, iii) exploring a suitable reference case against which to normalise hydrological performance, iv) identifying ways to define sustainable rates of water extractions against which to evaluate the sustainability of external water extraction, v) adding evaluation of water-related energy and nutrient flows, and vi) enabling assessment of climate change implications.

5. Conclusions

We took a big-picture perspective to the evaluation of the urban water performance, by extending an urban water metabolism evaluation framework to assess urban systems within three Australia city-regions. To test how it can inform water management decisions, we evaluated these regions under current conditions and after the implementation of example water sensitive interventions. Water performance was evaluated in terms of the external water efficiency and the hydrological performance of the urban system.

Adapting the evaluation framework to a large urban scale required the development of new methods for i) defining the system boundaries using land use data; and ii) for estimating natural hydrological flows by combining the land use data with hydrological flow partitioning (HFP) factors. The method can be conducted relatively rapidly, and could be used by water managers and urban planners for strategic evaluation. It is transferrable to other city-regions if spatial land use data are available, along with data for water supply and wastewater generation that can be aligned to the defined urban system boundaries.

We found that application of the evaluation framework to urban systems at the larger urban scale is suited to initial screening of the type and scale of water sensitive interventions needed to achieve desired water performance objectives, and to guide the establishment of performance objectives and targets. The values of the improved framework have been the broader urban perspective (to align with and inform strategic urban and water planning processes), and the systematic definition of the urban system boundary (to enable consistent comparisons between urban systems and over time). The framework can enable a better understanding of the role and impact of water sensitive interventions in achieving water performance objectives for urban systems within city-regions.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.watres.2018.01.070.

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